

MIT Sudoku Open 2026

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		T			O	K	U	
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O		U			I		K	
	K					T		U

Instruction Booklet

Schedule

10:00 – 10:30	Round 1: Classics	450 Points
10:40 – 11:20	Round 2: Variants I	500 Points
11:30 – 12:15	Round 3: Variants II	895 Points
13:00 – 13:45	Round 4: Feed-Forward	1000 Points
14:00 – 14:45*	Team Round 1: Line them up!	800 Points
15:00 – 15:45*	Team Round 2: More Interlocked Stratagems	900 Points

Competition Rules & Comments

Scoring & Bonuses

Full points will be awarded only for fully and corrected solved puzzles. By graders' discretion, partial credit of 90% and 80% will be given for puzzles with 1 or 2 minor mistakes, respectively. Some rounds may also have their own instructions on partial credit.

A bonus of 10 points for every full minute remaining will be awarded to any competitor who correctly solves all puzzles in an individual round. A partial 60% bonus can be awarded if one puzzle is incorrectly solved, under the condition that the puzzle is solved completely or almost completely and the competitor may have believed their solution to be correct. In case of doubt, decision will be made with graders' discretion.

A bonus of 40 points for every full minute remaining will be awarded to any team who correctly solves all puzzles in a round, with the same schematic as the individual bonus system above.

To declare completion of a round, a competitor must close their booklet and clearly state "finished" and raise their arm with the booklet. Same applies with team rounds.

Dishonesty Statement

Please no cheating everyone, you all are literally here on a Saturday doing sudoku for the sake of enjoying puzzles, why would you cheat at this, it makes no sense whatsoever.

Acknowledgements

Special thanks to MIT, since we would be solving sudoku outdoors without their indoor facilities.

Individual Round 1: Classics

30 Minutes – 450 Points

1.1 Classic Sudoku.....	20 Points
1.2 Classic Sudoku.....	30 Points
1.3 Classic Sudoku.....	25 Points
1.4 Classic Sudoku.....	50 Points
1.5 Classic Sudoku.....	30 Points
1.6 Classic Sudoku.....	25 Points
1.7 Classic Sudoku.....	15 Points
1.8 Classic Sudoku.....	20 Points
1.9 Classic Sudoku.....	15 Points
1.10 Classic Sudoku.....	15 Points
1.11 Classic Sudoku.....	15 Points
1.12 Classic Sudoku.....	15 Points
1.13 Classic Sudoku.....	65 Points
1.14 Classic Sudoku.....	55 Points
1.15 Classic Sudoku.....	55 Points

1.1 – 1.15: Classic Sudoku

Rules: Place the digits 1-9 (or 1-6) in each empty cell in the grid such that each row, column, and marked 3x3 box (or 2x3 box) contains each digit exactly once. [Online Solving](#)

	2						1	
4		6				7		3
	8			1			5	
			8		2			
		7				3		
			6		4			
	6			5			2	
2		5				4		1
	9						8	

5	2	3	7	4	9	8	1	6
4	1	6	2	8	5	7	9	3
7	8	9	3	1	6	2	5	4
6	3	1	8	7	2	5	4	9
8	4	7	5	9	1	3	6	2
9	5	2	6	3	4	1	7	8
1	6	8	4	5	3	9	2	7
2	7	5	9	6	8	4	3	1
3	9	4	1	2	7	6	8	5

Individual Round 2: Variants I

40 Minutes – 500 Points

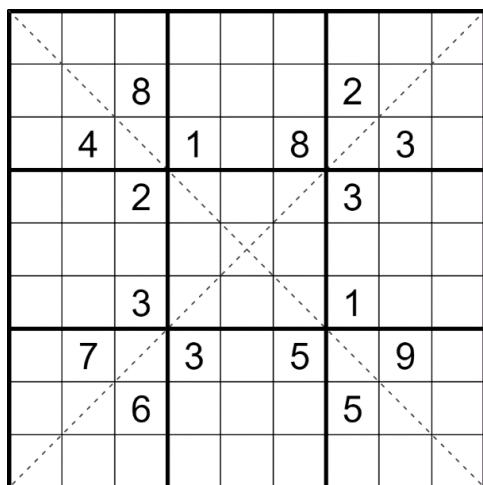
2.1 Antidiagonal Sudoku.....	10 Points
2.2 Antidiagonal Sudoku.....	35 Points
2.3 Antiknight Sudoku	15 Points
2.4 Antiknight Sudoku	65 Points
2.5 Odd/Even Sudoku	10 Points
2.6 Odd/Even Sudoku	35 Points
2.7 Outside Sudoku	20 Points
2.8 Outside Sudoku	65 Points
2.9 Sandwich Sums Sudoku	15 Points
2.10 Sandwich Sums Sudoku	50 Points
2.11 Skyscrapers Sudoku	20 Points
2.12 Skyscrapers Sudoku	110 Points
2.13 Thermo Sudoku.....	10 Points
2.14 Thermo Sudoku.....	40 Points

2.1 – 2.2: Antidiagonal Sudoku

10+35 Points

Rules: Classic sudoku rules apply. Each main diagonal contains exactly three distinct digits (two for 6x6 grids).

[Online Solving](#)



2.3 – 2.4: Antiknight Sudoku

15+65 Points

Rules: Apply Classic Sudoku rules. Cells that can be reached by a knight step (chess) cannot contain the same digit. [Online Solving](#)

1								3
	2						6	
		3				1		
			4		8			
				5				
			9		6			
		2				7		
	4						8	
3								9

1	6	9	5	4	2	8	7	3
8	2	5	3	7	1	9	6	4
4	7	3	6	8	9	1	5	2
5	9	6	4	2	8	3	1	7
2	1	4	7	5	3	6	9	8
7	3	8	9	1	6	4	2	5
6	5	2	8	9	4	7	3	1
9	4	1	2	3	7	5	8	6
3	8	7	1	6	5	2	4	9

2.5 – 2.6: Odd/Even Sudoku

10+35 Points

Rules: Classic sudoku rules apply. Digits in circles must be odd and digits in squares must be even. [Online Solving](#)

5				3			8	
	6				2			9
					1		4	
9								5
	2		9					
2			6				5	
	9			1				7

5	7	9	1	3	6	4	8	2
1	6	8	4	7	2	5	3	9
3	4	2	8	9	5	6	7	1
7	8	6	2	5	1	9	4	3
9	3	1	7	4	8	2	6	5
4	2	5	9	6	3	7	1	8
8	5	4	3	2	7	1	9	6
2	1	7	6	8	9	3	5	4
6	9	3	5	1	4	8	2	7

20+65 Points

Online Solving

2

4 1 6 3

9 3 8 7 5 1 7

4 6 4 3 9 6 8 5 9 4 3 5 8

1 3 4 9 6 8 5 7 9 4 3 5 8

1 7 9 3 2 1 7 2

4 8

8

15+50 Points

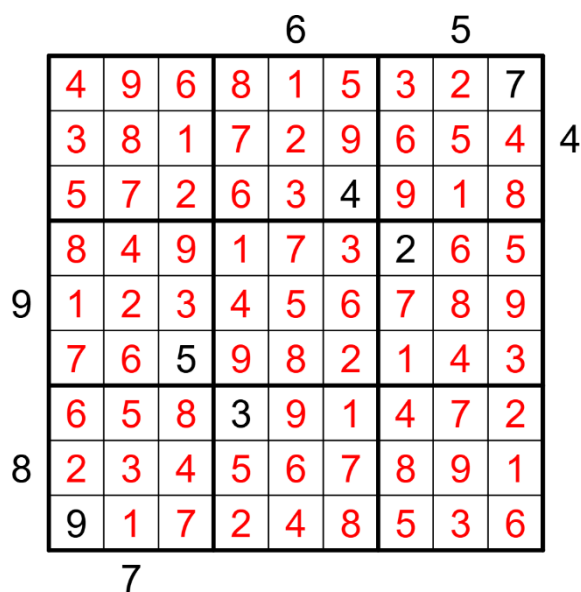
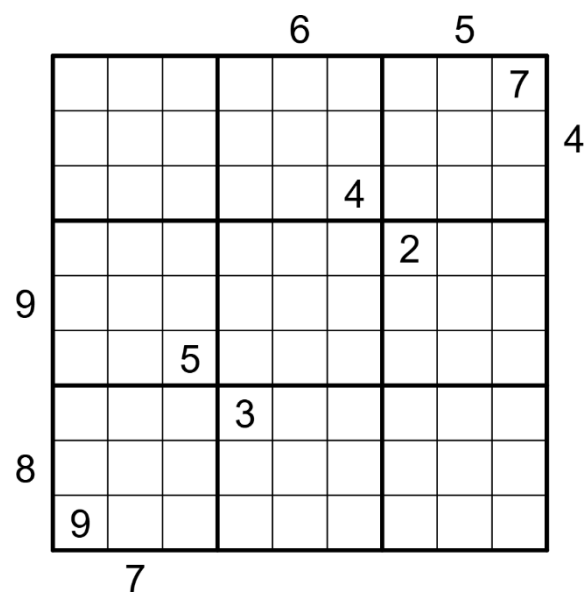
Online Solving

	20	20	0	22	14	22	20	30	2
8	3	7	4	1	8	9	5	2	6
29	2	1	8	3	6	5	7	9	4
35	9	5	6	2	7	4	3	8	1
7	8	6	3	4	9	7	1	5	2
28	5	2	1	8	3	6	4	7	9
7	7	4	9	5	2	1	8	6	3
5	1	3	2	9	5	8	6	4	7
21	6	9	5	7	4	3	2	1	8
2	4	8	7	6	1	2	9	3	5

2.11 – 2.12: Skyscrapers Sudoku

20+110 Points

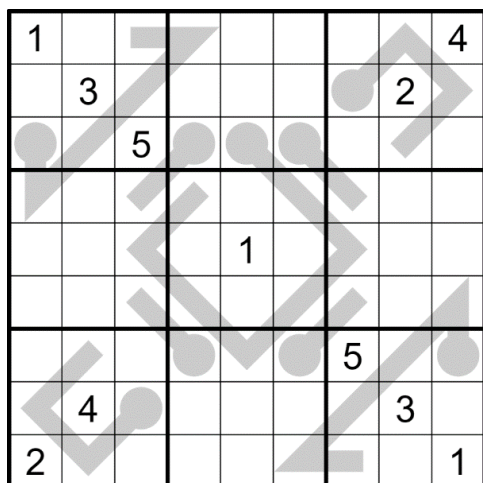
Rules: Classic sudoku rules apply. Each digit represents the height of a building. The clues outside the grid indicate the number of buildings visible from the corresponding direction. A taller building will hide any shorter buildings behind it. [Online Solving](#)



2.13 – 2.14: Thermo Sudoku

10+40 Points

Rules: Classic sudoku rules apply. Some thermometer shapes are placed in the grid. Digits are strictly increasing from the round bulb of each thermometer to each flat end. [Online Solving](#)



Individual Round 3: Variants II

45 Minutes – 895 Points

3.1 Deficit Sudoku.....	35 Points
3.2 Surplus Sudoku.....	40 Points
3.3 Battleships Sudoku.....	65 Points
3.4 Battleships Sudoku.....	200 Points
3.5 Consecutive Clone Sudoku.....	55 Points
3.6 Consecutive Clone Sudoku.....	50 Points
3.7 Elimination Sudoku.....	10 Points
3.8 Elimination Sudoku.....	100 Points
3.9 Liar Classic Sudoku.....	15 Points
3.10 Liar Classic Sudoku.....	125 Points
3.11 Little Killer Sudoku.....	35 Points
3.12 Little Killer Sudoku.....	95 Points
3.13 Wheels Sudoku.....	30 Points
3.14 Wheels Sudoku.....	40 Points

3.1: Deficit Sudoku

35 Points

Rules: Place a number from 1-N in each empty cell in the grid such that each row and column contains each number exactly once, and each marked region contains each number at most once. [Online Solving](#)

	1	2	5	6	4	
	5				7	
	7				3	
	4				2	
	6	3	4	2	5	

4	3	6	7	5	1	2
7	1	2	5	6	4	3
2	5	4	3	1	7	6
6	7	1	2	4	3	5
3	4	5	6	7	2	1
1	6	3	4	2	5	7
5	2	7	1	3	6	4

3.2: Surplus Sudoku

40 Points

Rules: Classic sudoku rules apply. Place a number from 1-N in each empty cell in the grid such that each row and column contains each number exactly once, and each marked N+1 cell region contains each number at least once. [Online Solving](#)

			3			
	3	6	2	5	4	
	5				7	
4	2				6	3
	1				3	
	4	1	6	3	5	
			5			

1	6	4	3	7	2	5
7	3	6	2	5	4	1
3	5	2	1	6	7	4
4	2	5	7	1	6	3
5	1	7	4	2	3	6
2	4	1	6	3	5	7
6	7	3	5	4	1	2

3.3 – 3.4: Battleships Sudoku

65+200 Points

Rules: Classic sudoku rules and Battleships rules apply. Clues outside the grid indicate the number of ship segments in its row/column. Ships may be rotated, and different ships cannot be placed in adjacent cells that share an edge or corner. Given digits are also considered seas, so no ship should be placed on them. [Online Solving](#)

									4
		3				6			
	1	9	4		2	5	7		2
		5				8			
									3
		8				3			
	9	6	3		1	7	5		2
		4				9			
									4
4	2		3		3		4		
	5	3	7	1					
3	9	4		5	6	7			
2	3		1	8		2	5		
5		6		7			9		

6	4	2	5	3	7	1	8	9	4
7	5	3	1	8	9	6	2	4	
8	1	9	4	6	2	5	7	3	2
3	6	5	9	2	4	8	1	7	
9	7	1	8	5	3	2	4	6	3
4	2	8	7	1	6	3	9	5	
2	9	6	3	4	1	7	5	8	2
5	3	4	2	7	8	9	6	1	
1	8	7	6	9	5	4	3	2	4
	5	3	7	1					
3	9	4		5	6	7			
2	3		1	8		2	5		
5		6		7			9		

3.5 – 3.6: Consecutive Clone Sudoku

55+50 Points

Rules: Classic sudoku rules apply. In each shaded region of the same shape, the numbers placed in corresponding cells of the regions must be consecutive. [Online Solving](#)

6				1	7			
				3				
		4						
								5
1	3						2	9
8								
						4		
				9				
			6	2				1

6	5	8	4	1	7	2	9	3
9	1	2	8	3	5	7	4	6
3	7	4	2	6	9	5	1	8
2	4	7	9	8	1	3	6	5
1	3	5	7	4	6	8	2	9
8	6	9	3	5	2	1	7	4
5	9	6	1	7	8	4	3	2
4	2	1	5	9	3	6	8	7
7	8	3	6	2	4	9	5	1

3.7 – 3.8: Elimination Sudoku

10+100 Points

Rules: Apply Classic Sudoku rules. If there is a digit N in a cell with an arrow, the digit N cannot appear in a direction the arrow points at. [Online Solving](#)

	5	7						
1		8	9					
6	2		5	3				
	6	3		1	8			
		1	4		3	6		
			6	5		3	1	
				9	6		3	4
					2	7		9
						2	8	

4	5	7	2	8	1	9	6	3
1	3	8	9	6	7	5	4	2
6	2	9	5	3	4	8	7	1
9	6	3	7	1	8	4	2	5
5	9	1	4	2	3	6	9	7
7	4	2	6	5	9	3	1	8
2	7	5	8	9	6	1	3	4
8	1	6	3	4	2	7	5	9
3	9	4	1	7	5	2	8	6

15+125 Points

	7		6		8		4	
2		9		7		5		6
	6		3		2		9	
8				3				4
		1		1		3		
6				4				2
	5		4		9		1	
9		8		1		4		5
	1		5		3		6	

3	7	5	6	9	8	2	4	1
2	8	9	1	7	4	5	3	6
1	6	4	3	5	2	7	9	8
8	2	7	9	3	1	6	5	4
5	4	1	2	6	7	3	8	9
6	9	3	8	4	5	1	7	2
7	5	6	4	2	9	8	1	3
9	3	8	7	1	6	4	2	5
4	1	2	5	8	3	9	6	7

35+95 Points

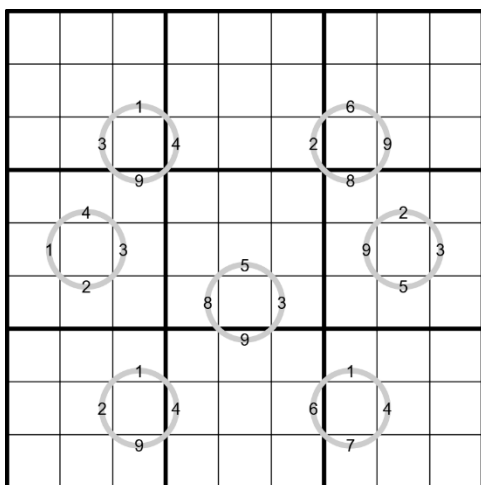
4	1	2	7	6	8	9	3	5
6	7	3	9	4	5	1	2	8
9	5	8	3	2	1	4	6	7
8	6	5	2	1	4	3	7	9
7	9	4	6	5	3	2	8	1
3	2	1	8	9	7	6	5	4
5	8	9	4	3	2	7	1	6
2	4	7	1	8	6	5	9	3
1	3	6	5	7	9	8	4	2

3.13 – 3.14: Wheels Sudoku

30+40 Points

Rules: Classic sudoku rules apply. The digits on the circles have to be placed in the same circular order in the four cells that are touched by the circle. The circles may have to be rotated to the correct position by 90, 180, or 270 degrees.

[Online Solving](#)



2	5	6	8	3	7	9	1	4
4	9	1	2	6	5	8	7	3
7	3	8	4	1	9	5	2	6
8	4	9	5	2	1	6	3	7
1	7	3	6	9	4	2	8	5
6	2	5	3	7	8	4	9	1
3	8	4	7	5	2	1	6	9
5	1	7	9	8	6	3	4	2
9	6	2	1	4	3	7	5	8

Individual Round 4: Feed Forward

45 Minutes – 1000 Points

4.1.1 Extra Regions Sudoku (6x6)	20 Points
4.1.2 Diagonal Sudoku (6x6)	20 Points
4.1.3 Renban Sudoku (6x6)	20 Points
4.1.4 Parity Regions Sudoku (6x6)	20 Points
4.2.1 Irregular Sudoku.....	80 Points
4.2.2 Scattered Sudoku.....	50 Points
4.2.3 Sudo-Kurve.....	30 Points
4.2.4 Toroidal Sudoku.....	110 Points
4.3.1 Killer Sudoku	60 Points
4.3.2 Arrow Sudoku.....	100 Points
4.3.3 X-Sums Sudoku.....	110 Points
4.3.4 Mathrax Sudoku	110 Points
4.4.1 Consecutive Pairs Sudoku.....	50 Points
4.4.2 Kropki Pairs Sudoku.....	70 Points
4.4.3 XV Pairs Sudoku.....	60 Points
4.4.4 Inequality Sudoku.....	90 Points

This round consists of 4 different cycles of 4 puzzles each, giving a total of 16 puzzles. Each cycle of 4 puzzles forms a progression of $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A \dots$, where each solved Sudoku provides information for the next sudoku. Without that information, the next puzzle may or may not have a unique solution.

These cells are marked with letters, with the first Sudoku containing the letters A1, A2 (carried onto the second puzzle) and D1, D2 (carried over from the fourth/last puzzle), and the second puzzle containing the letters A1, A2, B1, B2, and so on.

3 of the cycles contain 9x9 puzzles, while 1 contain 6x6 puzzles. You can also see that the 4 cycles all have different themes.

- Cycle 1: Extra Sets (6x6)
 - Extra Regions, Diagonal, Renban, Parity Regions
- Cycle 2: Geometry
 - Irregular, Scattered, Sudo-Kurve, Toroidal
- Cycle 3: Math
 - Killer, Arrow, X-Sums, Mathrax
- Cycle 4: Neighbors
 - Consecutive Pairs, Kropki Pairs, XV Pairs, Greater Than

This round is on the more difficult end, so partial credit will be given for 9x9 grids, in which each correctly completed 3x3 grid yields 10% of puzzle points.

4.1.1: Extra Regions Sudoku

20 Points

Rules: Classic Sudoku rules apply. Each orthogonally connected shaded region contains the digits 1-6. [Online Solving](#)

	1			2	
2					5
			3		
		4			
5					6
	3			4	

4	1	5	6	2	3
2	6	3	4	1	5
1	2	6	3	5	4
3	5	4	1	6	2
5	4	1	2	3	6
6	3	2	5	4	1

4.1.2: Diagonal Sudoku

20 Points

Rules: Classic Sudoku rules apply. Each diagonal contains the digits 1-6. [Online Solving](#)

				2	
					3
1	2				
2		1			
	4	5			

5	6	3	4	2	1
4	1	2	6	5	3
3	5	6	2	1	4
1	2	4	3	6	5
2	3	1	5	4	6
6	4	5	1	3	2

4.1.3: Renban Sudoku

20 Points

Rules: Classic sudoku rules apply. Orthogonally connected shaded regions (grey lines in example) must contain a set of distinct, consecutive digits in any order. [Online Solving](#)

	1			2	
5			1		4
				5	
	5				
3		5			1
	4			3	

6	1	4	3	2	5
5	3	2	1	6	4
2	6	1	4	5	3
4	5	3	2	1	6
3	2	5	6	4	1
1	4	6	5	3	2

4.1.4: Parity Regions Sudoku

20 Points

Rules: Classic Sudoku rules apply. Each grey region contains digits of the same parity. [Online Solving](#)

			3		
					2
					1
3					
4					
		1			

1	4	2	3	6	5
6	3	5	1	4	2
2	5	4	6	3	1
3	1	6	2	5	4
4	2	3	5	1	6
5	6	1	4	2	3

4.2.1: Irregular Sudoku

80 Points

Rules: Each row, column, and outlined region must include the digits 1-9 once, with no repeats. [Online Solving](#)

					5			
3						9		
	1	4						
	6	5			8			7
				2				
7			8			6	5	
						7	1	
		9						2
			5					

9	4	7	6	8	5	2	3	1
3	8	6	2	1	7	9	4	5
2	1	4	9	5	6	3	7	8
4	6	5	3	9	8	1	2	7
5	3	8	7	2	1	4	9	6
7	9	1	8	3	2	6	5	4
8	5	2	4	6	9	7	1	3
6	7	9	1	4	3	5	8	2
1	2	3	5	7	4	8	6	9

4.2.2: Scattered Sudoku

50 Points

Rules: Irregular Sudoku rules apply, with the 9 shaded cells forming an additional 9-cell region. [Online Solving](#)

				3				
			9		1			
		1				6		
	5		3	6	4		8	
1			6		9			7
	8		1	7	5		6	
		5				1		
			5		6			
				2				

6	9	4	2	3	7	5	1	8
5	6	8	9	4	1	7	2	3
3	2	1	7	5	8	6	4	9
9	5	7	3	6	4	2	8	1
1	3	2	6	8	9	4	5	7
4	8	9	1	7	5	3	6	2
7	4	5	8	9	2	1	3	6
2	7	3	5	1	6	8	9	4
8	1	6	4	2	3	9	7	5

30 Points

A 10x10 grid with a 3x3 sub-grid in the center. The sub-grid has numbers 4, 2, 8, 6, 3, 2, 5, 4, 9, 1 in its cells. The outer grid has numbers 3, 9, 8, 6, 1, 6, 2, 3 in its cells. Lines connect the sub-grid to the outer grid.

110 Points

	2	7	1	9	8	6	5	
1	5						8	9
2								7
6			4	7	2			8
7			9		6			2
3			7	4	9			5
8								4
5	7						3	6
	8	6	5	3	7	4	2	

4	2	7	1	9	8	6	5	3
1	5	3	6	2	4	7	8	9
2	4	9	8	1	3	5	6	7
6	3	5	4	7	2	1	9	8
7	1	8	9	5	6	3	4	2
3	6	2	7	4	9	8	1	5
8	9	1	3	6	5	2	7	4
5	7	4	2	8	1	9	3	6
9	8	6	5	3	7	4	2	1

4.3.1: Killer Sudoku

60 Points

Rules: Classic sudoku rules apply. The number at the top-left corner of each cage equals the sum of digits inside the cage. Digits do not repeat inside a cage. [Online Solving](#)

	2			4			3	
6		15			11			1
	9						6	
			9		2			
2		9			10			6
	13		7		4		8	
1		11			14			3
	3			7				9

9	2	8	1	4	5	6	3	7
6	5	15		7	8	3	11	
4	9		1	3	6	2	7	9
7			8	5	9	6	2	3
2	9	9		4	5	1	10	
3	13		6	1	7	8	4	5
8			7	2	3	9	1	4
1	4	11		9	2	5	14	
5	3			6	4	7	8	1

4.3.2: Arrow Sudoku

100 Points

Rules: Apply Classic Sudoku rules. Each digit placed in a cell with a circle must be the sum of the digits placed in the cells that the adjoining arrow passes through. Digits may repeat on arrow. [Online Solving](#)

			5		8			
				1				
	7						6	
		8				3		
	2						7	
				9				
		4		5				

5	8	9	3	7	2	1	4	6
1	6	7	5	4	8	2	9	3
2	3	4	9	1	6	5	8	7
4	7	5	2	3	9	8	6	1
9	1	8	7	6	4	3	2	5
3	2	6	8	5	1	9	7	4
8	5	3	6	9	7	4	1	2
6	9	1	4	2	5	7	3	8
7	4	2	1	8	3	6	5	9

110 Points

A 6x6 grid with numbers 11, 12, 40, 43, 21 at the top and 20 at the bottom. Numbers 29, 35, 25, 43, 6, 5 are on the left, and 38, 9, 25, 27, 28, 44 are on the right.

110 Points

4			3		2			7
	11+	9+		1			4/	
					8			
	E	9		2		4		O
			5	8	3			
		5		4		2		2
		18x	6				2-	
				7				
7			8		1			2

4	1	8	3	6	2	5	9	7
8	2	6	7	1	9	8	4	3
9	3	7	4	5	8	1	2	6
6	8	9	1	2	7	4	3	5
2	4	1	5	8	3	6	7	9
3	7	5	9	4	6	2	8	1
8	6	2	6	9	5	7	1	4
1	9	3	2	7	4	3	5	8
7	5	4	8	3	1	9	6	2

50 Points

Online Solving

5	4	7	1	9	6	3	8	2
2	3	6	7	8	5	4	9	1
8	9	1	2	3	4	5	6	7
7	8	2	9	5	1	6	3	4
6	5	3	8	4	2	7	1	9
9	1	4	3	6	7	8	2	5
1	2	5	6	7	8	9	4	3
3	7	8	4	1	9	2	5	6
4	6	9	5	2	3	1	7	8

70 Points

Online Solving

9	8	7	6	5	4	3	2	1
5	6	2	3	1	8	9	4	7
4	3	1	2	9	7	6	8	5
3	4	9	1	2	6	5	7	8
1	5	8	7	4	3	2	6	9
7	2	9	9	8	5	4	1	3
2	1	3	8	6	9	7	5	4
8	7	5	4	3	2	1	9	6
6	9	4	5	7	1	8	3	2

4.4.3: XV Pairs Sudoku

60 Points

Rules: Classic sudoku rules apply. A “V” symbol between two cells indicates that the digits in those two cells sum to 5. An “X” symbol between two cells indicates that the digits in those two cells sum to 10. Not all “X”s and “V”s are given. [Online Solving](#)

		x			x				
									v
x					x	x			
		1			v				x
			2						
				3					v
v		x							
		x			4				
							5		
								x	
		x							

6	3	x	7	4	1	x	9	8	5	2
2	4	5	6	8	7	1	9	3		v
x	8	9	1	5	x	x	2	7	6	x
5	6	9	2	7	1	3	4	8		
4	8	x	2	9	3	5	6	1	7	
v	1	7	x	3	8	6	4	5	2	5
9	2	8	3	4	6	5	7	1		
7	5	4	1	x	9	8	2	x	3	6
3	1	x	6	7	5	2	4	8	9	

4.4.4: Inequality Sudoku

90 Points

Rules: Classic Sudoku rules apply. If there is an inequality relation marked between two cells, then the digits placed in the cells should obey the relation. [Online Solving](#)

	1				2			3	
	4				5			6	
	7				8			9	

2	5	9	4	>	3	7	8	>	1	6
4	1	6	>	5	2	8	<	9	3	7
3	8	>	7	9	6	>	1	2	5	>
6	9	1	8	>	7	3	4	>	2	5
7	4	8	>	2	5	9	>	1	6	3
5	3	>	2	1	4	<	6	7	8	<
8	>	2	4	3	<	9	5	6	<	7
1	7	3	<	6	8	4	<	5	9	2
9	6	>	5	7	1	<	2	3	4	8

Team Round 1: Line them Up

45 Minutes* – 800 Points

5.1 Arrow Sudoku.....	100 Points
5.2 German Whispers Sudoku.....	100 Points
5.3 No Bulb Thermo Sudoku	100 Points
5.4 Palindrome Sudoku.....	100 Points
5.5 Region Sum Lines Sudoku.....	100 Points
5.6 Renban Sudoku.....	100 Points
5.7 Same Parity Lines Sudoku.....	100 Points
5.8 Arithmetic Sequences Sudoku.....	100 Points

*Approximate time; team rounds will be run for max(45 minutes, 2nd submission + 5 minutes)

5.1-8: Line Them Up

8 * 100 Points

Each of the eight grids contain digits, but are missing their line components. There are eight other separate grids with line components, but missing digits. You must match up each numbered grid with each lined grid, and solve the resulting puzzles. The lined grids can be rotated, but not reflected. You may write answers on either of the grids.

Variants & Rules

Arrow Sudoku

Rules: Apply Classic Sudoku rules. Place a circle at one end of each line – the digit placed in the circle must be the sum of all other digits on the line. [Online Solving](#)

5	8	9	3	7	2	1	4	6
1	6	7	5	4	8	2	9	3
2	3	4	9	1	6	5	8	7
4	7	5	2	3	9	8	6	1
9	1	8	7	6	4	3	2	5
3	2	6	8	5	1	9	7	4
8	5	3	6	9	7	4	1	2
6	9	1	4	2	5	7	3	8
7	4	2	1	8	3	6	5	9

German Whispers Sudoku

Rules: Classic sudoku rules apply. Adjacent digits along the line(s) must have a difference of at least 5. [Online Solving](#)

	7			9				
3				1				2
			8				7	
		1				3		
	9				6			
4				2				1
			7				8	
		7				4		
	8				4			

8	7	2	4	5	9	1	3	6
3	4	9	6	1	7	8	5	2
5	1	6	8	3	2	9	7	4
6	2	1	5	7	8	3	4	9
7	9	3	1	4	6	5	2	8
4	5	8	9	2	3	7	6	1
2	3	4	7	9	1	6	8	5
9	6	7	2	8	5	4	1	3
1	8	5	3	6	4	2	9	7

No Bulb Thermo Sudoku

Rules: Classic sudoku rules apply. Digits placed along each line must form a strictly increasing sequence from one end or the other. [Online Solving](#)

2	6		8					
7						2		
							3	
	9							
		6						7
				4		2	8	

2	6	4	8	7	3	5	1	9
7	3	9	5	4	1	2	8	6
5	1	8	6	2	9	7	3	4
3	4	7	1	9	6	8	5	2
6	8	1	7	5	2	9	4	3
9	5	2	4	3	8	6	7	1
8	9	3	2	1	7	4	6	5
4	2	6	3	8	5	1	9	7
1	7	5	9	6	4	3	2	8

Palindrome Sudoku

Rules: Classic sudoku rules apply. Numbers along shaded lines must be the same forwards and backwards. [Online Solving](#)

		9						1
		8					6	
5	6	7			4			
			1			6		
				5				
		2			9			
			8			3	4	5
	8						2	
7						1		

2	4	9	7	6	8	5	3	1
3	1	8	9	2	5	4	6	7
5	6	7	3	1	4	8	9	2
8	9	3	1	7	2	6	5	4
4	7	1	6	5	3	9	2	8
6	5	2	4	8	9	7	1	3
1	2	6	8	9	7	3	4	5
9	8	4	5	3	1	2	7	6
7	3	5	2	4	6	1	8	9

Region Sum Lines Sudoku

Rules: Classic sudoku rules apply. For each line, digits on the line have an equal sum N within each box it passes through. If a line passes through the same box more than once, each individual segment of such a line within that box sums to N separately. [Online Solving](#).

4					2			
						5		
							3	
								6
			8					
3								
	8							
		6						
			4					3

4	6	8	3	5	2	9	1	7
9	1	3	6	4	7	5	2	8
2	5	7	1	9	8	6	3	4
8	4	5	2	1	9	3	7	6
6	2	9	7	8	3	1	4	5
3	7	1	5	6	4	2	8	9
1	8	4	9	3	6	7	5	2
7	3	6	8	2	5	4	9	1
5	9	2	4	7	1	8	6	3

Renban Sudoku

Rules: Classic sudoku rules apply. Gray lines must contain a set of distinct, consecutive digits in any order. [Online Solving](#)

			2		1			
		9						
6			8				9	
				4				
	4		3					
1								

1	4	6	3	9	8	2	7	5
2	9	5	6	7	1	3	4	8
7	3	8	4	2	5	1	6	9
8	5	7	9	1	3	4	2	6
4	6	1	5	8	2	7	9	3
3	2	9	7	6	4	8	5	1
5	7	4	8	3	6	9	1	2
9	1	3	2	5	7	6	8	4
6	8	2	1	4	9	5	3	7

Same Parity Lines Sudoku

Rules: Classic sudoku rules apply. On each grey line, either all digits are odd or all are even. [Online Solving](#)

	4		2	8		5	
2		5		4			8
			6			1	
	5				9		7
1		2				4	
	2			1			
3			6		1		5
	1		3	9		7	

9	4	1	2	3	8	7	5	6
2	6	5	1	4	7	3	9	8
8	3	7	6	9	5	4	1	2
6	5	3	4	1	2	9	8	7
4	7	8	9	5	3	2	6	1
1	9	2	8	7	6	5	4	3
7	2	4	5	8	1	6	3	9
3	8	9	7	6	4	1	2	5
5	1	6	3	2	9	8	7	4

Arithmetic Sequences Sudoku

Rules: Classic sudoku rules apply. On each grey line, the digits form an arithmetic sequence, starting at one end and ending at the other. [Online Solving](#)

						4	
							8
7							
	3						

2	7	8	9	5	1	3	4	6
3	6	1	2	4	7	5	9	8
9	5	4	3	8	6	7	2	1
1	2	3	4	6	9	8	5	7
4	8	7	5	3	2	1	6	9
5	9	6	7	1	8	4	3	2
6	4	9	8	7	3	2	1	5
7	1	5	6	2	4	9	8	3
8	3	2	1	9	5	6	7	4

Team Round 2: More Interlocked Strategems

45 Minutes – 900 Points

6.1 Coded Pointing Evens Sudoku	150 Points
6.2 Hidden Dihedral Shape Sudoku.....	150 Points
6.3 Killer Sudoku	150 Points
6.4 Love Sudoku	150 Points
6.5 Partial 159 Sudoku	150 Points
6.6 Pencilmarks Irregular Sudoku	150 Points

*Approximate time; team rounds will be run for max(45 minutes, 2nd submission + 5 minutes)

Setup

This round consists of six different 9x9 grids with different variants. In addition, cells that appear in the same relative grid position in the six puzzles must be different. There is exactly one solution that fulfills both the mutual exclusive rule and the regular Sudoku variant rules for the four puzzles.

Variants & Rules

6.1: Coded Pointing Evens Sudoku

150 Points

Rules: Apply Classic Sudoku rules. Clues outside the grid indicate the number of even digits in the direction that it points. There are a series of letters across the grid that each represents a value. No two letters can have the same value. [Online Solving](#)

	1	2	3					
	4	5	6					
	2	3	4					
V								
E	R	Y						

6	1	2	3	4	5			
3	4	5	6	2	1			
5	2	3	4	1	6			
7	6	4	2	5	3			
4	5	6	1	3	2			
2	3	1	5	6	4			

6.2: Hidden Dihedral Shape

150 Points

Rules: Classic sudoku rules apply. There are some numbered shapes that must be put into the grid. Shapes can be rotated (in 45 degree increments) AND reflected. Different shapes may not intersect each other or share cells. [Online Solving](#)

	1			2			3	
	4			5			6	
	7			8			9	

				1	2	3	3	4
		5	2	6	6	5	6	5
6	2	5			3	7	5	7
		5	7	1	9	5	9	8
1	7	5						

2	5	9	7	3	6	4	1	8
4	1	6	9	2	8	7	3	5
7	3	8	1	4	5	9	2	6
3	2	7	4	6	9	5	8	1
8	4	1	2	5	7	3	6	9
6	9	5	8	1	3	2	7	4
9	8	3	5	7	1	6	4	2
5	7	2	6	8	4	1	9	3
1	6	4	3	9	2	8	5	7

				1	2	3	3	4
		5	2	6	6	5	6	5
6	2	5			3	7	5	7
		5	7	1	9	5	9	8
1	7	5						

6.3: Killer Sudoku

150 Points

Rules: Classic sudoku rules apply. The number at the top-left corner of each cage equals the sum of digits inside the cage. Digits do not repeat inside a cage. [Online Solving](#)

	2			4			3	
6		15			11			1
	9						6	
			9		2			
2		9			10			6
	13		7		4		8	
1		11			14			3
	3			7			9	

9	2	8	1	4	5	6	3	7
6	5	7	8	3	9	2	4	1
4	1	3	6	2	7	9	5	8
7	8	5	9	6	2	3	1	4
2	9	4	5	1	3	7	8	6
3	6	1	7	8	4	5	2	9
8	7	2	3	9	1	4	6	5
1	4	9	2	5	6	8	7	3
5	3	6	4	7	8	1	9	2

6.4: Love Sudoku

150 Points

Rules: Classic sudoku rules apply. There are two pairs of “lovers” in this grid (1-2 & 7-8, only 1-2 in this example). Each pair of lovers is a pair of digits in the puzzle, in which every instance of one digit will be adjacent (orthogonally or diagonally) to only one instance of its lover. [Online Solving](#)

	2			1	
1		6			3
4			5		2
	5			3	

3	2	5	6	1	4
1	4	6	2	5	3
5	1	4	3	2	6
2	6	3	1	4	5
4	3	1	5	6	2
6	5	2	4	3	1

6.5: Partial 159 Sudoku

150 Points

Rules: Classic sudoku rules apply. If a digit N is placed in the shaded cells in columns 1, 5, or 9, then in the Nth column in that row there is a 1, 5, or 9, respectively. All shaded cells are present. (Example only includes 1 and 5). [Online Solving](#).

				2	
					3
			3		
2					
	3				

1	6	3	4	2	5
5	2	4	6	1	3
3	1	6	5	4	2
4	5	2	3	6	1
2	4	5	1	3	6
6	3	1	2	5	4

6.6: Pencilmarks Irregular Sudoku

150 Points

Rules: Irregular sudoku rules apply. Additionally, some cells include pencilmarks, which list the only possibilities for the value that the cell may take. [Online Solving](#)

16					34
		25	56		
		12	15		
		45	34		
		56	23		
23					12

6	5	3	1	2	4
2	4	5	6	3	1
4	3	2	5	1	6
1	2	4	3	6	5
5	1	6	2	4	3
3	6	1	4	5	2